

Derek Wang

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TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, TypeScript, SQL

Frameworks: React, Next.js, Node.js, Express, FastAPI

Libraries/Tools: PyTorch, OpenCV, SQLite, PostgreSQL, MongoDB, Docker, Git

Robotics: ROS2, Foxglove, Fusion 360

Certifications: Docker Foundations Professional

EXPERIENCE

Autonomous Self Driving Software Engineer

Sept 2025 – Present

WATonomous

Waterloo, ON, Canada

- Architected and deployed ROS2 navigation stack (costmaps, A* planning, Pure Pursuit) enabling autonomous rover path following in dynamic environments (hills, obstacles, holes), simulating Mars environment
- Led YOLO11s model retraining, improving detection mAP from 30% to 80% across 3,448 labeled images via hyperparameter tuning and augmentation redesign
- Optimized YOLO11s inference to 30 FPS on rover hardware via ONNX export and onnxslim optimization

Autonomy Developer

Aug 2025 – Feb 2026

Waterloo Aerial Robotics Group

Waterloo, ON, Canada

- Developed Python hardware-in-the-loop UAV simulation using Ardupilot Mission Planner, Pixhawk flight controller and MAVLink protocol for the 2026 UAS Student Competition
- Eliminated GPS teleportation bugs upon injection and achieved smooth trajectory interpolation at 24 FPS via periodic position updates, drone speed control, position thresholding, and trajectory smoothing algorithms

Robotics Software Programmer – VEX Robotics Competition

Sept 2024 – June 2025

Checkmate Robotics Club

Markham, Ontario

- Reduced autonomous robot positioning error to $\pm 0.5\text{cm}$ by implementing wheel odometry, distance sensors, and IMU for position estimation, and PID control with trapezoidal motion profiling algorithms for trajectory control
- Qualified for 2025 VEX World Championships (top 10% globally) and placed 5th/80 at Ontario Provincials

PROJECTS

Silhouette | *Node.js, Next.js, Express, MongoDB, Gemini, Overshoot.ai SDK*

- Developed a real-time AI fashion advisor with live camera input at NexHacks, earning a VC referral to continue development of Proof of Concept for Merian Brothers
- Designed and delivered a MERN full-stack application for real-time outfit analysis and product recommendations
- Orchestrated Overshoot.ai and Gemini vision APIs, designing prompt pipelines that manage async inference, rate limits, and style constraints to deliver personalized recommendations in under 2s

Hierarchical Reasoning Retrieval | *Node.js, Express, SQLite, Gemini*

- Designed hierarchical LLM document retrieval system traversing heading trees instead of flat vector similarity
- Decreased LLM token usage by up to 80% and optimized retrieval by engineering an SQLite pipeline that chunks markdown into document trees using bottom-up summarization, keyword pre-filtering, and tree search
- Implemented beam search traversal over document heading trees for structured document search

Autonomous Vehicle Control System | *C++, ROS2, Docker, CMake, Foxglove*

- Built a C++ LiDAR perception pipeline generating real-time occupancy costmaps at 0.1m resolution in Foxglove
- Developed ROS2 autonomous navigation stack with Docker, validated 100% obstacle avoidance in 50+ simulations
- Simulated robot navigation and visualized perception outputs in Foxglove for debugging and testing

EDUCATION

University of Waterloo

Expected Apr 2030

Bachelor of Applied Science in Computer Engineering (Co-op)

GPA: 3.7

- Relevant Coursework:** Linear Algebra, Fundamentals of Programming